

Homework 8G: Church-Turing Thesis and Turing-Acceptability

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You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments. Four bonus points if you use L^AT_EX, and our template. You may collaborate with other students, but any written work should be your own. **Many of these problems have two sentence answers if you can apply the theorems proved in class.**

1. Find a problem that satisfies each of the following conditions:

- (a) (6 points) It is NP-hard but not in NP.

Answer: If a game of chess were to be played on an $n \times n$ board, then determining if a particular player is able to force a win (i.e. is able to win despite what moves the opponent makes), is NP-hard but not in NP. We find this to be true since determining if a forced win is possible is exponential time as a result of the exponential number of moves both players are able to make.

- (b) (6 points) It is in NP but not NP-complete.

Answer: The problem of factoring an integer is in NP but is not NP-complete. We observe if we are given an integer along with its factors (not including 1), then we are able to verify the correctness of the factor (via multiplication) in poly-time. Therefore, we find the problem is NP. If this problem were NP-complete, then an implication may be $NP = \text{complement-NP}$, which has not been proven (and I will not be the one to do so in this simple example).

2. We know that the Hamiltonian Path Problem is NP-complete and the Euler Circuit Problem is solvable in polynomial time so it is in P. Using these two facts, prove whether the following problems are in P or in NP-complete by applying polynomial-time reductions:

- (a) (4 points) TSP Problem

Answer: We wish to reduce the Hamiltonian Path Problem to TSP in poly-time. Given a graph G , we wish to construct a complete graph by assigning edges existing in the original graph with a weight of 1. Edges that do not exist in the original graph will have an edge weight of 2. We observe a TSP tour with cost equal to the number of vertices exists if and only if there exists a Hamiltonian Path in the original graph. Therefore, since there exists a poly-time reduction from the Hamiltonian Path Problem to TSP, and we can verify the solution in poly-time, we find TSP to be NP-Complete.

- (b) (4 points) Integer Programming Problem

Answer: We wish to reduce the Hamiltonian Path problem to the Integer Programming Problem. We wish to create binary variables x_{ij} for each edge i, j in

the graph. An assignment of $x_{ij} = 1$ indicates the edge is part of the path. We then define linear constraints such that (1) each vertex has exactly one incoming and outgoing edge and (2) the selected edges must form a single path, visiting each vertex. Since this reduction from Hamiltonian to Integer Programming is done in poly-time, and since verifying the assignments can be done in poly-time, we find the Integer Programming Problem is NP-Complete.

- (c) (4 points) String Construction Problem: Given n 2-length strings $t_1, t_2, \dots, t_n$, construct a string s of length $n+1$ such that the multiset $\{s_1s_2, s_2s_3, \dots, s_ns_{n+1}\}$ is equal to the given multiset $\{t_1, t_2, \dots, t_n\}$. For example, if $t_1 = ab$ and $t_2 = bc$, a valid solution is $s = abc$.

Answer: We wish to take each 2-length string t_i and treat it as a directed edge. Therefore, we let $t_i = ab$ represent a directed edge t_i connecting a to b . We observe that the desired string s of $len = n+1$ must have its consecutive 2-letter substrings equal to the multiset. Regarding graphs, this is the same as finding a trail that uses every edge exactly once. We find this is equivalent to the Euler Circuit Problem which is solvable in poly-time. Therefore, since we can reduce the String Construction Problem to the Euler Circuit Problem, we find both problems to be in P since they are solvable in poly-time.

- (d) (4 points) Multiset Split Problem: Given a multiset of n even-sized integer sets $s_1, s_2, \dots, s_n$ (where each $|s_i|$ is even), find a split for each $s_i = p_i \cup q_i$ such that $|p_i| = |q_i| = \frac{1}{2}|s_i|$, and the union of all p_i 's equals the union of all q_i 's. For example, $s_1 = \{1, 1\}$, $s_2 = \{2, 2\}$, then one solution is $p_1 = q_1 = \{1\}$, $p_2 = q_2 = \{2\}$

Answer: We wish to reduce the Multiset Split Problem to the Hamiltonian Path Problem. Given a directed graph G , we reduce to the Hamiltonian Path Problem as follows: (1) for each edge $e = (u, v)$, we include a 2-element set such that $s_e = \{u, v\}$ and (2) include an extra set s^* that contains two copies of each vertex. This will ensure every vertex appears in both p and q .

In any valid split, we observe every vertex appears at least once in the left and right parts. We observe in a Hamiltonian Cycle, each vertex appears exactly once as the tail of one edge and once as the head of another edge. With our two rules, as defined above, we find the union over all p 's and q 's are identical. If a balanced split is possible, then the choices force a permutation of the vertices which correspond to a Hamiltonian cycle.

Since we have reduced from the Multiset Split problem to the Hamiltonian Graph Problem, and since the Multisplit problem can be solved in poly-time, we find the Multiset problem is NP-complete.

3. (12 points) Let M be an $f(n)$ space bounded machine and consider its computation on some word w . Prove that there exists a CNF formula ϕ of size $O(f(n)^2)$ such that for any strings C_i, C_{i+1} , We have that $\phi(C_i, C_{i+1}) = 1$ if and only if C_i, C_{i+1} are configurations of M on w and $C_i \vdash C_{i+1}$ via δ of M .

Answer: We wish to construct a CNF formula ϕ such that it is satisfied by the encoding of two configurations C_i and C_{i+1} if and only if $C_i \vdash C_{i+1}$ according to the

transition function δ of the TM M .

We observe each configuration is encoded as a string of $len = f(n)$. Every cell in the string encodes a tape symbol, and one cell also encodes the machine's current state together with the head's position. We wish to introduce Boolean variables to represent the content of each cell in both configurations C_i and C_{i+1} .

The transition function δ only affects the cell under the head and the cells to the left and right (i.e. its neighbors). Therefore, we are able to construct ϕ with the following guidelines:

- (a) For cells not near the head, we add clauses that enforce the identity of the tape symbol in C_i and C_{i+1} . In other words, the symbol at cell j remains unchanged.
- (b) For the active cells (i.e. the head and its neighbors), we add clauses that abide by δ . These clauses ensure that if the head is at a specific cell in C_i , then the symbols in C_{i+1} reflect the transition defined by δ .

We observe there are $f(n)$ cells that we must check at each transition. Therefore, we find the total size to be $O(f(n) \cdot f(n) = O(f(n)^2)$.

Therefore, we find there exists a CNF formula ϕ of size $O(f(n)^2)$ such that $\phi(C_i, C_{i+1}) = 1$ if and only if C_i and C_{i+1} are valid configurations of M on input w , with $C_i \vdash C_{i+1}$ via the transition function δ .